

# Worksheet — Velocity Components Simulation

Simulation 13 · use with *Simulation - Velocity components.html* · A.1 Kinematics: velocity of a projectile — components, speed, direction,  $\Delta v = g\Delta t$ ; goes with video (or strobe-photo) analysis of a thrown ball

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

## How to use this worksheet

Open **Simulation - Velocity components.html** in a browser (it works offline). Set the launch, press  Launch or drag the **time probe**. The **velocity diagram** draws the velocities every 0.2 s from one point. ● **Record reading** saves the probe values.

**Part 1** before the lab ( $\approx 15$  min) · **Part 2** during the video analysis ( $\approx 15$  min) · **Part 3** homework ( $\approx 30$  min).

$v_x = u \cos \theta$  (constant) ·  $v_y = u \sin \theta - gt$  ·  $v = \sqrt{v_x^2 + v_y^2}$  ·  $\phi = \tan^{-1}(v_y/v_x)$ . Use  $g = 9.81 \text{ m s}^{-2}$ , up positive.

## Part 1 — Pre-lab: predict, then test

$u = 12 \text{ m s}^{-1}$  at  $50^\circ$  from the ground. Predict **before** you look, then test and explain.

| Question                                                               | My prediction (before) | What the simulation showed | Physics explanation |
|------------------------------------------------------------------------|------------------------|----------------------------|---------------------|
| 1. Where on the path is the speed smallest? What is it there?          |                        |                            |                     |
| 2. How does the velocity change from one 0.2 s vector to the next?     |                        |                            |                     |
| 3. Landing speed and direction compared with the launch (same height)? |                        |                            |                     |
| 4. Is there a point where the acceleration is zero?                    |                        |                            |                     |

## 5. Calculate the speed of the ball at the top of its path.

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## Part 2 — During the lab: video analysis

**Video.** Film a thrown ball next to a metre rule (or use a strobe photo). Read its position every 0.10 s. Average velocity between two frames =  $\Delta(\text{position}) \div \Delta t$ .

| t / s | x / m | y / m | $v_x = \Delta x / \Delta t$ / m s <sup>-1</sup> | $v_y = \Delta y / \Delta t$ / m s <sup>-1</sup> | at t = ... s (midpoint) |
|-------|-------|-------|-------------------------------------------------|-------------------------------------------------|-------------------------|
|       |       |       |                                                 |                                                 |                         |
|       |       |       |                                                 |                                                 |                         |
|       |       |       |                                                 |                                                 |                         |
|       |       |       |                                                 |                                                 |                         |
|       |       |       |                                                 |                                                 |                         |
|       |       |       |                                                 |                                                 |                         |

1. Simulation (12 m s<sup>-1</sup>, 50°): record the positions at t = 0.40 s and 0.60 s. Calculate the average  $v_y$  between them.

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2. At which time does the simulation's  $v_y$  equal your answer to Q1? Why that time?

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3. Record  $v_y$  every 0.2 s and fit a line on the Data tab.  $g =$  \_\_\_\_\_ m s<sup>-2</sup>

4.  $g$  from your video: \_\_\_\_\_ m s<sup>-2</sup>. Give one reason the two values differ.

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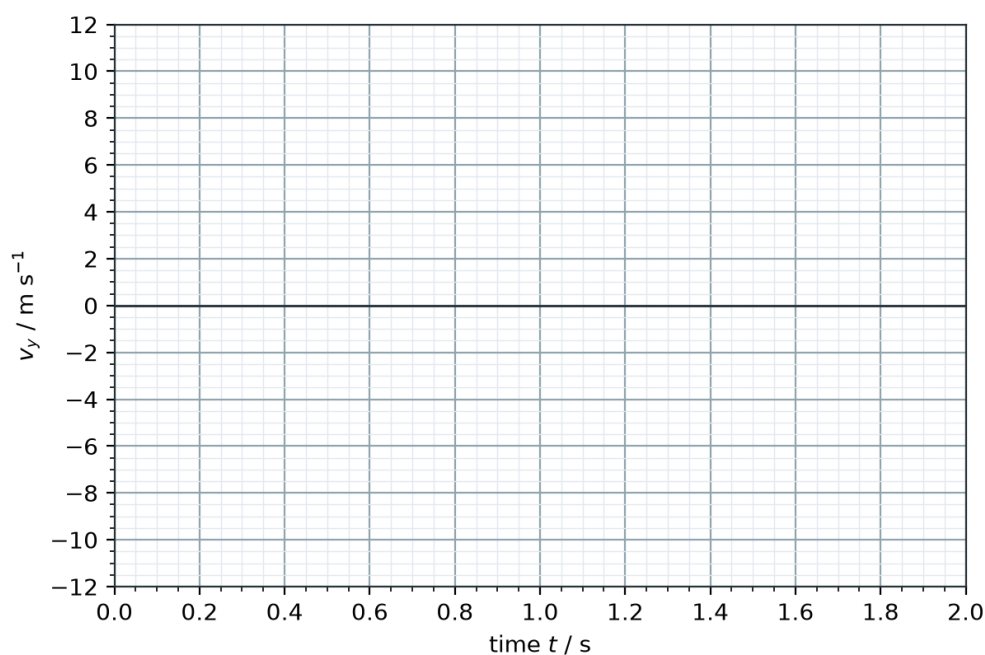
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## Part 3 — Homework: analyse

Simulation,  $12 \text{ m s}^{-1}$  at  $50^\circ$ : record  $v_y$  at each time.

| $t / \text{s}$          | 0.0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 | 1.2 | 1.4 | 1.6 | 1.8 |
|-------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| $v_y / \text{m s}^{-1}$ |     |     |     |     |     |     |     |     |     |     |

1. Plot  $v_y$  against  $t$  and draw the best-fit line. Gradient = \_\_\_\_\_ ; intercept = \_\_\_\_\_. What do they represent?



2. A ball is thrown from the ground at  $20 \text{ m s}^{-1}$ ,  $60^\circ$  above the horizontal. Calculate its speed 1.5 s after the throw.

3. At what angle to the horizontal is it moving at that moment? Draw the velocity triangle.

4. At what time is the ball moving at  $45^\circ$  below the horizontal?

**5. What is the speed of the ball when it is 10 m above the ground?**

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**6. Ball P is thrown at  $20 \text{ m s}^{-1}$  at  $30^\circ$ , ball Q at  $20 \text{ m s}^{-1}$  at  $70^\circ$ . When each is 5 m above the launch point, which is faster? Explain.**

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